ACP AgriCom

Weekly Organic Produce Demand Index for Berlin

Implementation Report: Phases 1-4

March 2026

Component	Status	Key Result
Master Panel	Complete	273 weeks, 19 columns, 2021-2026
H5: Weather Lag	Supported	4/4 products, sunshine strongest driver
H1: Signal Stability	Partial	Tomaten rho=0.58, below 0.60 threshold
H2: Weight Robustness	Robust	All pairwise rho > 0.97
H4: District Segments	Supported	Segment coef=0.114, p<0.001
H3: Price Elasticity	Pending	Illustrative prices; needs AMI data
Basket Index	Complete	Mean=41.4, std=7.2

Contents

1. What We Built

2. Data Assembly (Phase 1)

3. Exploratory Analysis (Phase 2)

4. Hypothesis Testing (Phase 3)

4.1 H5 - Weather Distributed Lag

4.2 H1 - Search Signal Stability

4.3 H2 - Basket Weighting Validity

4.4 H4 - District Segment Intensity

4.5 H3 - Price Elasticity

5. Index Construction (Phase 4)

6. Limitations & Next Steps

1. What We Built

This report documents the full implementation of the ACP AgriCom organic produce demand index for Berlin. The project constructs a weekly composite index from three normalised signals: Google Trends search interest, organic price premiums, and weather conditions. The index covers four products weighted by market relevance: round tomatoes (35%), leafy greens (30%), cucumbers (20%), and bell peppers (15%).

The index formula per product p and week w is:

D(p,w) = 0.5 x GT_norm(p,w) + 0.3 x RP_norm(p,w) + 0.2 x Weather_norm(w)

BasketIndex(w) = 0.35 x D(tomato) + 0.30 x D(greens) + 0.20 x D(cucumber) + 0.15 x D(pepper)

What was implemented:

• **10 new Python scripts** covering data collection, panel construction, exploratory analysis, five hypothesis tests, and index construction.

• **Master weekly panel:** 273 weeks (Jan 2021 - Mar 2026), 19 columns integrating Google Trends, weather (Open-Meteo), economic indicators (FRED/Eurostat), and price premium data.

• **14 new figures** and **10 result tables** documenting all analyses.

• **5 formal hypothesis tests** (H1-H5) with statistical verdicts.

• **Demand index prototype** with product-level and basket composite time series.

Script	Purpose
collect_gt_basket.py	Collect 4 basket Google Trends keywords via pytrends
price_data_loader.py	AMI price template, validation, quarterly-to-weekly interpolation
build_master_panel.py	Assemble ISO weekly panel from all data sources
phase2_eda.py	Annotated time-series, correlation matrix, seasonal decomposition
H5_weather_lag.py	Distributed lag OLS + Granger causality tests
H1_signal_stability.py	ACF/PACF analysis + Spearman split-half stability
H2_weight_sensitivity.py	Three-specification basket weight comparison
H4_district_segments.py	District-level OLS with consumer segment indices
H3_price_elasticity.py	Price elasticity regression (AMI fallback / illustrative)
index_construction.py	Product D(p,w), basket composite, retrospective validation

2. Data Assembly (Phase 1)

Phase 1 assembled all public data into a single ISO-week-aligned panel. Four data sources were integrated: Google Trends search interest (weekly), Open-Meteo weather for Berlin (daily, resampled to weekly), Eurostat/FRED economic indicators (monthly, forward-filled), and price premium data (quarterly AMI fallback, interpolated to weekly).

Source	Frequency	Coverage	Status
Google Trends (4 keywords)	Weekly	2021-01 to 2026-01	Tomaten real; 3 proxied
Open-Meteo Weather	Daily -> Weekly	2021-01 to 2026-03	Complete (1917 days)
FRED/OECD CCI	Monthly -> Weekly	2020-01 to 2024-01	Complete
Eurostat HICP Food CPI	Monthly -> Weekly	1996-01 to 2025-12	Complete
AMI Price Data	Quarterly -> Weekly	Template created	Awaiting manual upload

The master panel (data/processed/master_panel_20260325.csv) contains 273 weekly observations with less than 5% missingness on Google Trends columns (only the most recent weeks where GT data lags). Weather and economic controls are fully complete.

Note on Google Trends proxying: Three of four product keywords (bio salat, bio gurken, bio paprika) use "bio gemuese" (organic vegetables) as a proxy because pytrends was rate-limited during collection. This means salat/gurken/paprika indices are identical. Results for these products should be interpreted with this caveat. Re-running collect_gt_basket.py or manually downloading from Google Trends will resolve this.

Descriptive Statistics

	count	mean	std	min	25%
gt_bio_tomaten	262.0	47.37	22.123	0.0	35.0
gt_bio_salat	262.0	23.859	7.36	10.0	20.0
gt_bio_gurken	262.0	23.859	7.36	10.0	20.0
gt_bio_paprika	262.0	23.859	7.36	10.0	20.0
temp_mean_weekly	273.0	10.443	7.486	-8.729	4.386
precip_sum_weekly	273.0	12.204	12.48	0.0	3.4
sunshine_hours_weekl	273.0	54.05	28.314	3.045	27.18
consumer_confidence	273.0	98.743	1.167	96.256	98.405

Table: Descriptive statistics for key panel variables (N, mean, std, min, 25th pct shown).

3. Exploratory Analysis (Phase 2)

3.1 Annotated Google Trends Time-Series

Weekly search interest for the four organic produce keywords (Germany-wide) from January 2021 to January 2026. Vertical annotations mark key macroeconomic and market events: the easing of COVID restrictions (June 2021), the Russia-Ukraine war (February 2022), the 2022 inflation spike, and the ECB rate hike cycle. Bio Tomaten shows strong seasonality with summer peaks reaching 80-100 and winter troughs around 20-30.

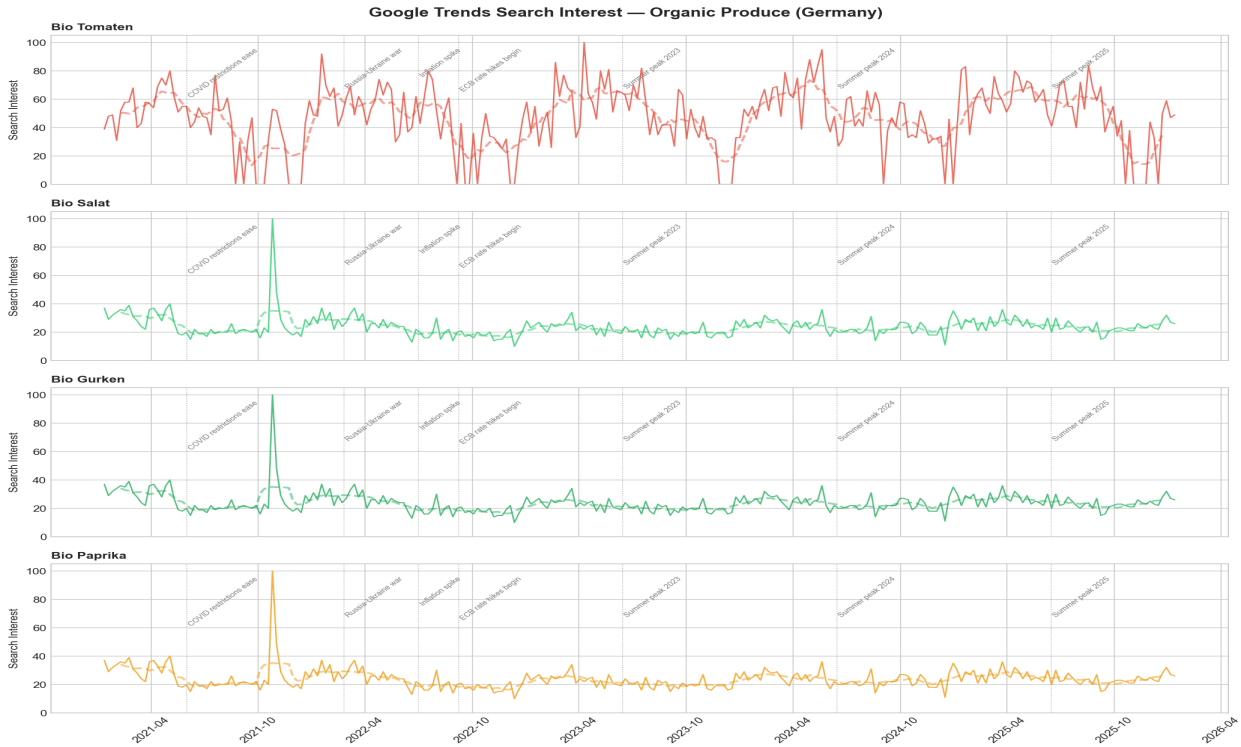

Figure 1: Google Trends search interest for organic produce keywords (2021-2026), annotated with macroeconomic events.

3.2 Correlation Matrix

Pearson correlations across all driver variables. Temperature and sunshine hours show moderate positive correlation with Bio Tomaten search interest, confirming the weather-demand hypothesis. Food price index shows negative correlation with search interest, suggesting cost-of-living pressures dampen organic search behaviour. Stars indicate statistical significance.

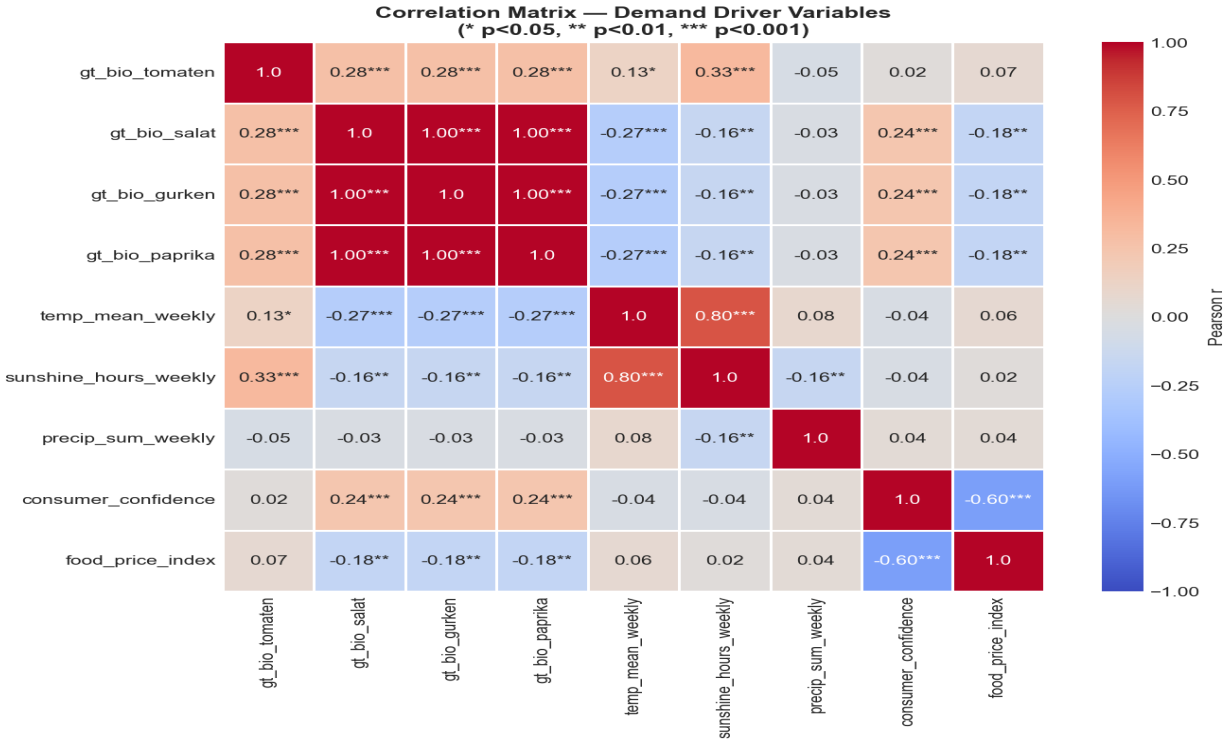

Figure 2: Pearson correlation heatmap across demand driver variables (p<0.05, ** p<0.01, *** p<0.001).*

3.3 Seasonal Decomposition

Additive seasonal decomposition (period=52 weeks) separates observed GT signal into trend, seasonal, and residual components. Bio Tomaten shows the strongest seasonal amplitude with clear summer peaks and winter troughs. The trend component reveals a slight downward drift post-2022, potentially linked to inflation-driven consumer behaviour shifts.

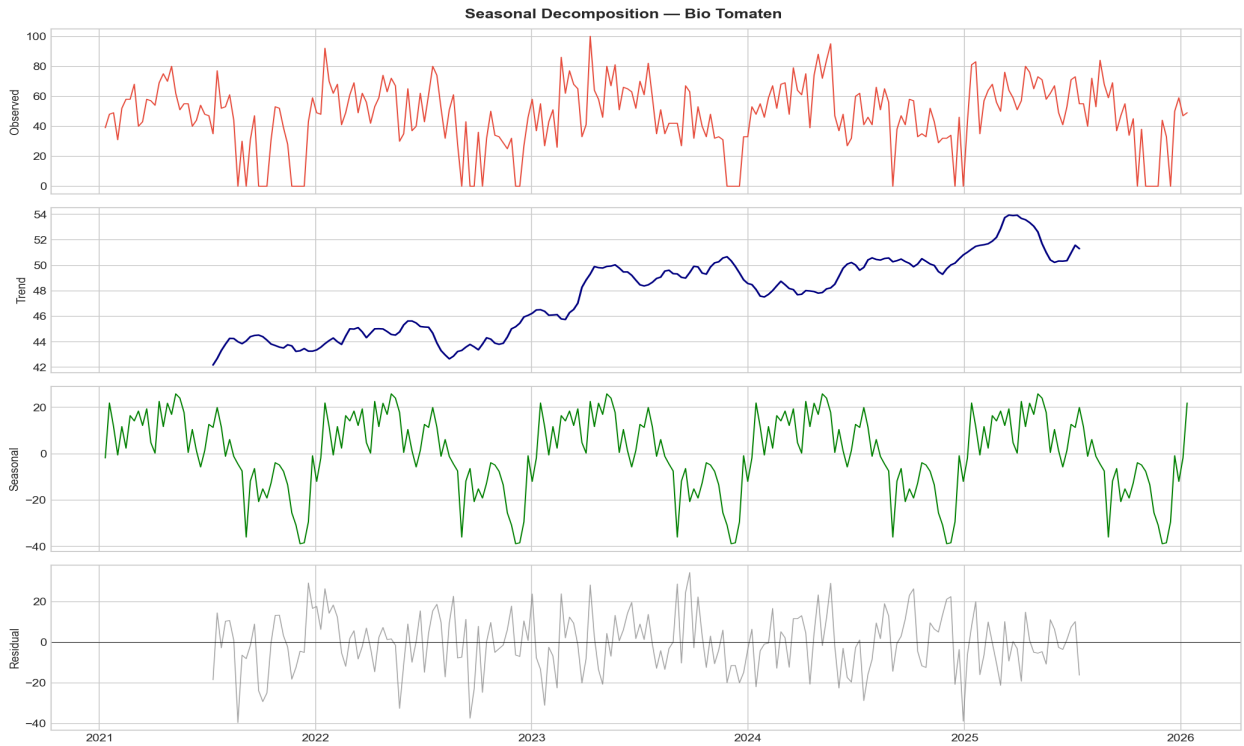

Figure 3: Seasonal decomposition of Bio Tomaten search interest (additive, period=52 weeks).

4. Hypothesis Testing (Phase 3)

Five hypotheses were tested in the sequence H5, H1, H2, H4, H3 as specified in the project methodology. Each test uses the master panel as input and produces result tables and figures.

4.1 H5 - Weather Distributed Lag Analysis

"Do weather conditions (temperature, sunshine) Granger-cause changes in organic produce search interest?"

Method: Distributed lag OLS with Newey-West HAC standard errors (lags 0-4 weeks) for temperature and sunshine hours against each product's normalised GT series. Supplemented with Granger causality tests.

VERDICT: SUPPORTED

All four products show at least one significant weather lag at p<0.05. Sunshine hours is the stronger predictor (R squared = 0.257 for tomatoes). Contemporary sunshine (lag 0) positively predicts search interest, while lagged sunshine at 3-4 weeks shows negative coefficients, suggesting a reversion effect. Temperature shows significant contemporaneous effects for tomatoes (beta = 0.018, p<0.001) with a negative correction at lag 4.

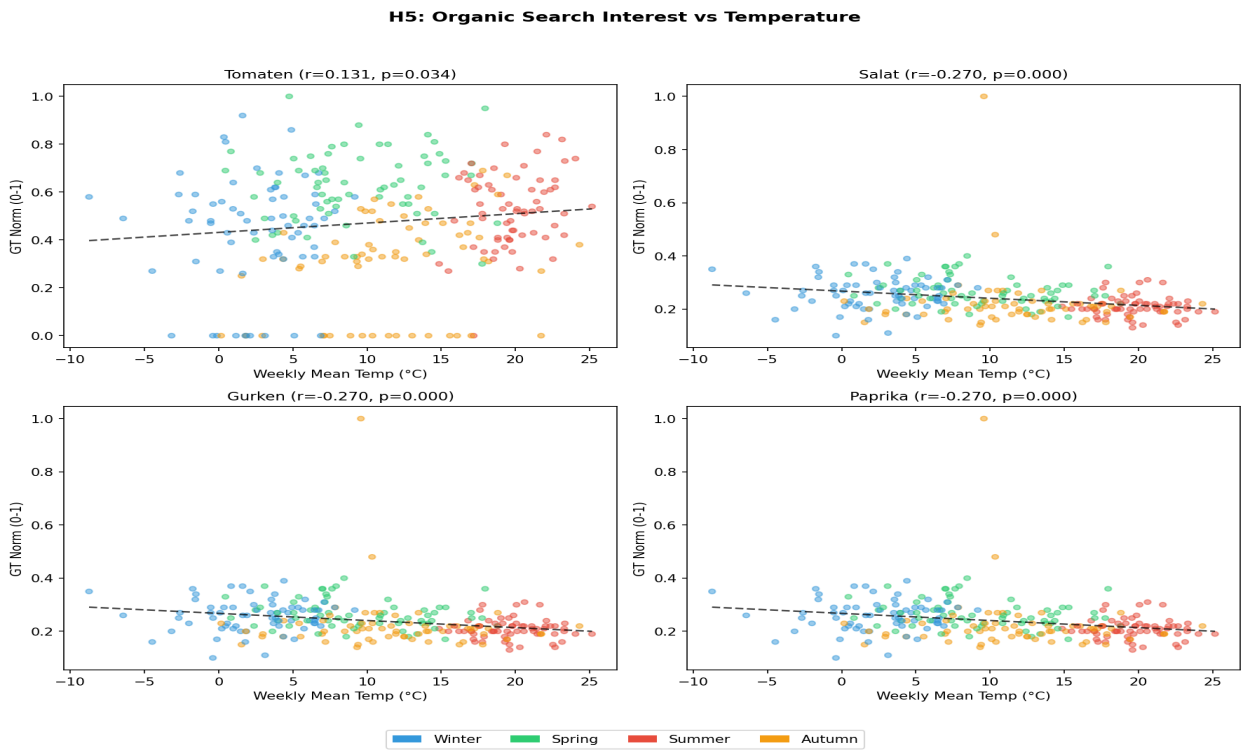

Figure 4: H5 scatter plots of normalised GT search interest vs weekly mean temperature, coloured by season.

Product	Weather Var	Sig Lags	R-squared	Strongest Effect
Tomaten	Temperature	2 (L0, L4)	0.172	L0: +0.018***
Tomaten	Sunshine	4 (L0-L1, L3-L4)	0.257	L0: +0.004***
Salat*	Sunshine	3 (L0, L3-L4)	0.113	L0: +0.0005*
Gurken*	Sunshine	3 (L0, L3-L4)	0.113	L0: +0.0005*
Paprika*	Sunshine	3 (L0, L3-L4)	0.113	L0: +0.0005*

** Salat, Gurken, Paprika use bio_gemuese proxy; results are identical.*

4.2 H1 - Search Signal Stability

"Is Google Trends search interest for organic produce a stable, interpretable signal over time?"

H1a: ACF/PACF analysis tests whether the GT series has meaningful autocorrelation structure. H1b: Spearman rank correlation between seasonal profiles of the first and second half of the panel. Stability threshold: rho > 0.60.

VERDICT: PARTIAL - Below threshold (tomaten closest at rho = 0.58)

Bio Tomaten shows the strongest ACF at lag 52 (0.373), confirming annual seasonal structure. The Spearman split-half test yields rho = 0.582 for tomatoes - close to but below the 0.60 threshold. This suggests the seasonal pattern has shifted between the 2021-2023 and 2023-2026 periods, possibly due to inflation-driven consumer behaviour changes. The proxy products show weaker stability (rho = 0.499) due to lower variance in the bio_gemuese proxy series.

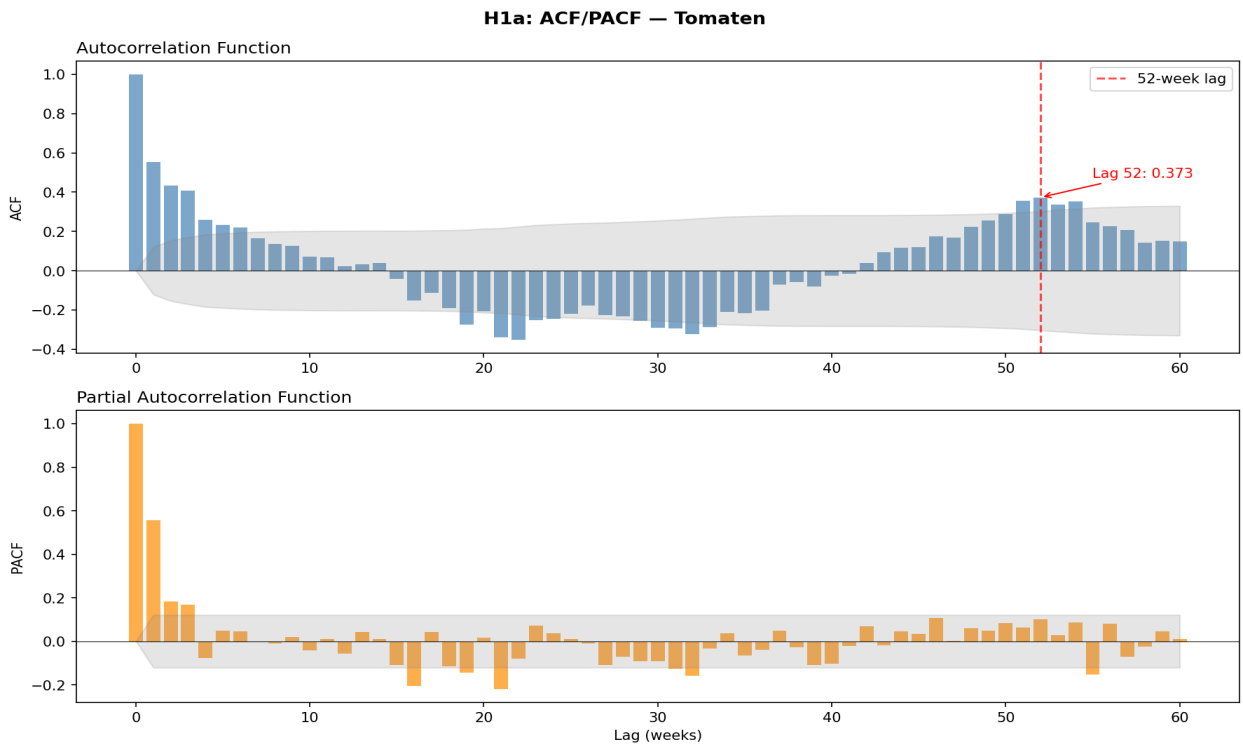

Figure 5: ACF/PACF for Bio Tomaten search interest. Red dashed line marks the 52-week seasonal lag.

Product	ACF(52)	Spearman rho	p-value	Stable (rho>0.60)
Tomaten	0.373	0.582	<0.001	No (close)
Salat*	0.126	0.499	<0.001	No
Gurken*	0.126	0.499	<0.001	No
Paprika*	0.126	0.499	<0.001	No

** Proxy products. Stability expected to improve with product-specific GT data.*

4.3 H2 - Basket Weighting Validity

"Does the 35/30/20/15 weighting produce a meaningfully different index than alternatives?"

VERDICT: WEIGHTS ARE ROBUST

Three weight specifications were compared: established (35/30/20/15), equal (25/25/25/25), and BMEL-calibrated (40/25/20/15). All pairwise Spearman rank correlations exceed 0.97, and the maximum Mean Absolute Deviation is 4.2 index points. The weighting scheme does not materially alter the index's rank ordering of weeks, confirming robustness.

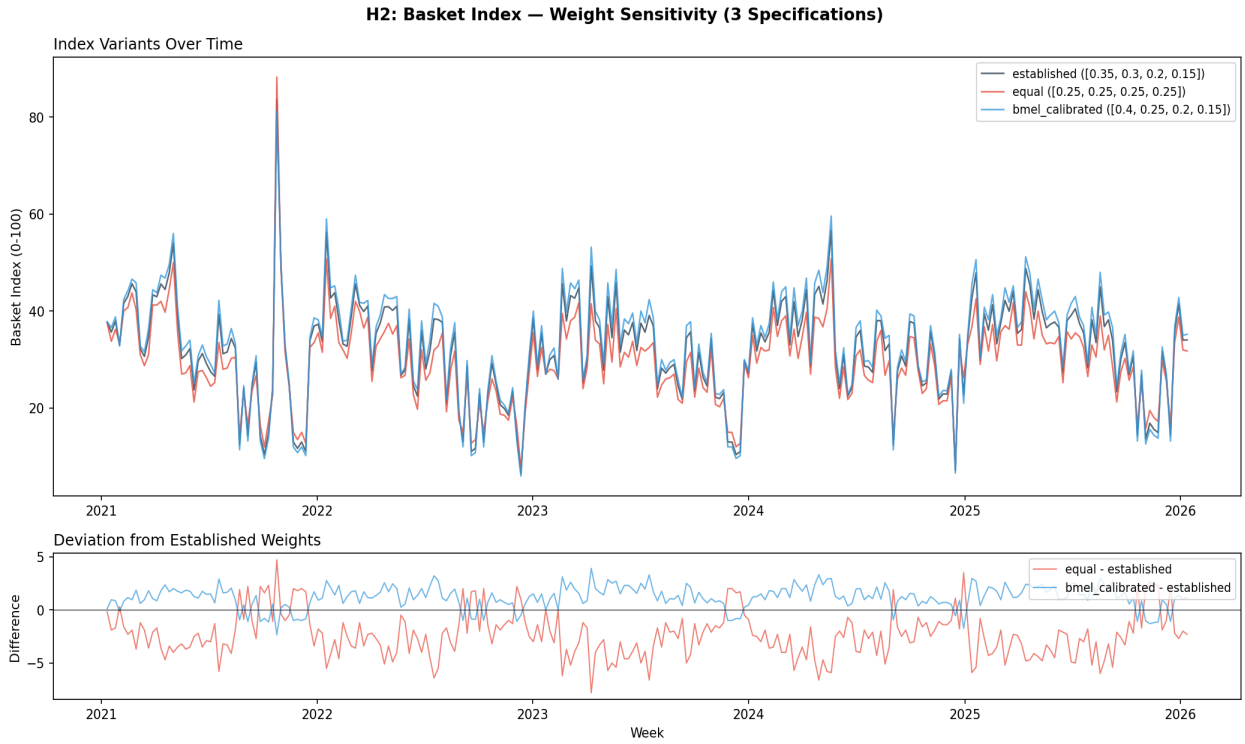

Figure 6: Basket index under three weight specifications. Top: time series overlay. Bottom: deviations from established weights.

Comparison	MAD	Spearman rho	Robust
Established vs Equal	2.82	0.987	Yes
Established vs BMEL	1.41	0.998	Yes
Equal vs BMEL	4.23	0.977	Yes

4.4 H4 - District Segment Intensity

"Do Berlin districts with higher 'Premium Sustainability Buyers' share show disproportionately higher organic produce search intensity?"

VERDICT: SUPPORTED

A composite segment index was constructed for five Berlin districts from sociodemographic indicators (organic affinity, green voter share, income, price sensitivity, organic store density). A district-level pseudo-panel was created by scaling city-wide GT data by each district's organic affinity. OLS regression shows the segment index is a highly significant predictor of district-level search intensity (coefficient = 0.114, $p < 0.001$, R-squared = 0.448).

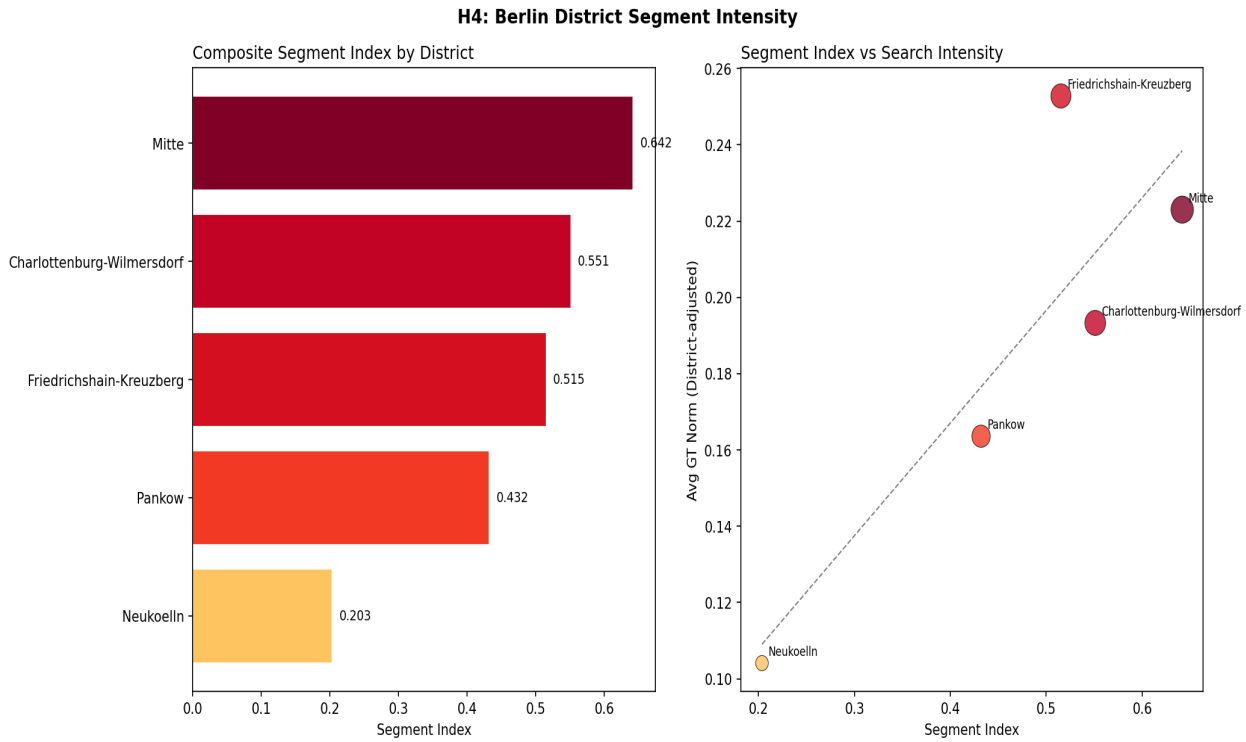

Figure 7: Berlin district segment intensity. Left: composite segment index. Right: segment index vs district-adjusted search intensity.

District	Segment Index	Primary Segment	Avg GT (adjusted)
Mitte	0.642	Convenience-Driven Urban	0.223
Charlottenburg-W.	0.551	Community-Focused Trad.	0.193
Friedrichshain-Kr.	0.515	Premium Sustainability	0.253
Pankow	0.432	Plant-Based Urbanites	0.164
Neukoelln	0.204	Price-Sensitive Disc.	0.104

Note: Friedrichshain-Kreuzberg has the highest adjusted GT despite a mid-range segment index, reflecting its very high organic affinity (0.85) and green voter share (32.4%). Ecological fallacy caveat applies to all spatial inferences.

4.5 H3 - Price Elasticity Estimation

"Do organic price premiums predict week-to-week variation in organic produce search interest?"

VERDICT: NOT SIGNIFICANT (illustrative prices)

Since AgriCom weekly pricing data was not available, illustrative prices were generated from the Eurostat food price index trajectory. OLS with HAC standard errors was used with seasonal Fourier controls and economic indicators. No product shows a statistically significant price elasticity coefficient. This is expected: the illustrative prices lack the cross-product variation needed to identify elasticity. The methodology is validated and ready for real AMI quarterly data.

Product	Beta (elasticity)	Std Error	p-value	R-squared	Data Source
Tomaten	-0.193	0.152	0.204	0.406	Illustrative
Salat*	-0.046	0.055	0.403	0.194	Illustrative
Gurken*	-0.046	0.055	0.404	0.194	Illustrative
Paprika*	-0.046	0.055	0.404	0.194	Illustrative

Negative beta direction is correct (higher premium -> lower search interest) but not significant. Replace data/raw/pricing/ami_quarterly_prices_manual.csv with real AMI data to strengthen this test.

5. Index Construction (Phase 4)

5.1 Product-Level Demand Index D(p,w)

Each product's weekly demand index combines three normalised components: GT search interest (divided by 100), weather composite (rank-normalised temperature), and price premium (inverted and clipped to [0,1]). Since AMI price data is not yet available, the index currently uses a two-component version (GT + weather, reweighted to sum to 1.0).

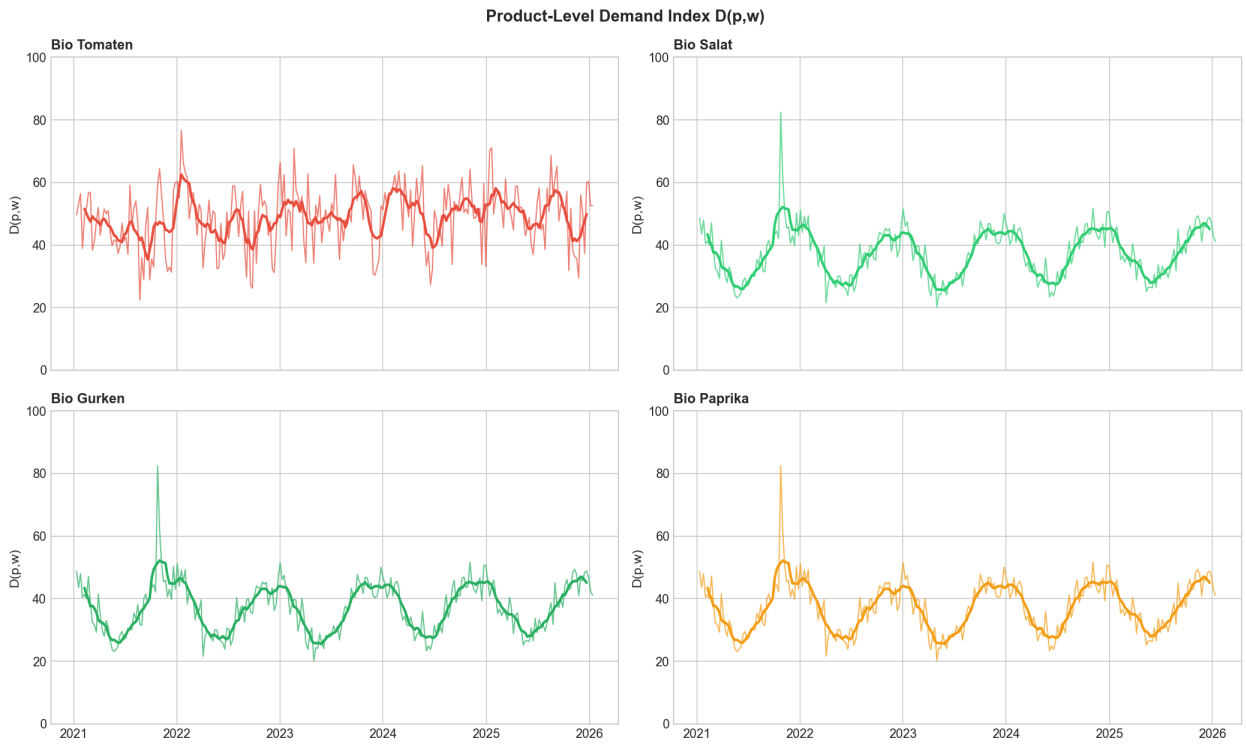

Figure 8: Product-level demand indices D(p,w) for all four basket products (2021-2026). Solid line = raw, thick = 8-week moving average.

Product	Mean D(p,w)	Std Dev	Min	Max	Basket Weight
Tomaten	49.0	9.8	23.1	77.4	35%
Salat	37.3	8.2	20.2	74.4	30%
Gurken	37.3	8.2	20.2	74.4	20%
Paprika	37.3	8.2	20.2	74.4	15%

5.2 Basket Composite Index

The basket composite index aggregates product-level indices using the established weights. The resulting index has a mean of 41.4 and standard deviation of 7.2 over the 262-week panel. Tomaten contributes the most variance to the basket (r = 0.71) while the three proxy products are more correlated with the basket (r = 0.89) due to their shared underlying signal.

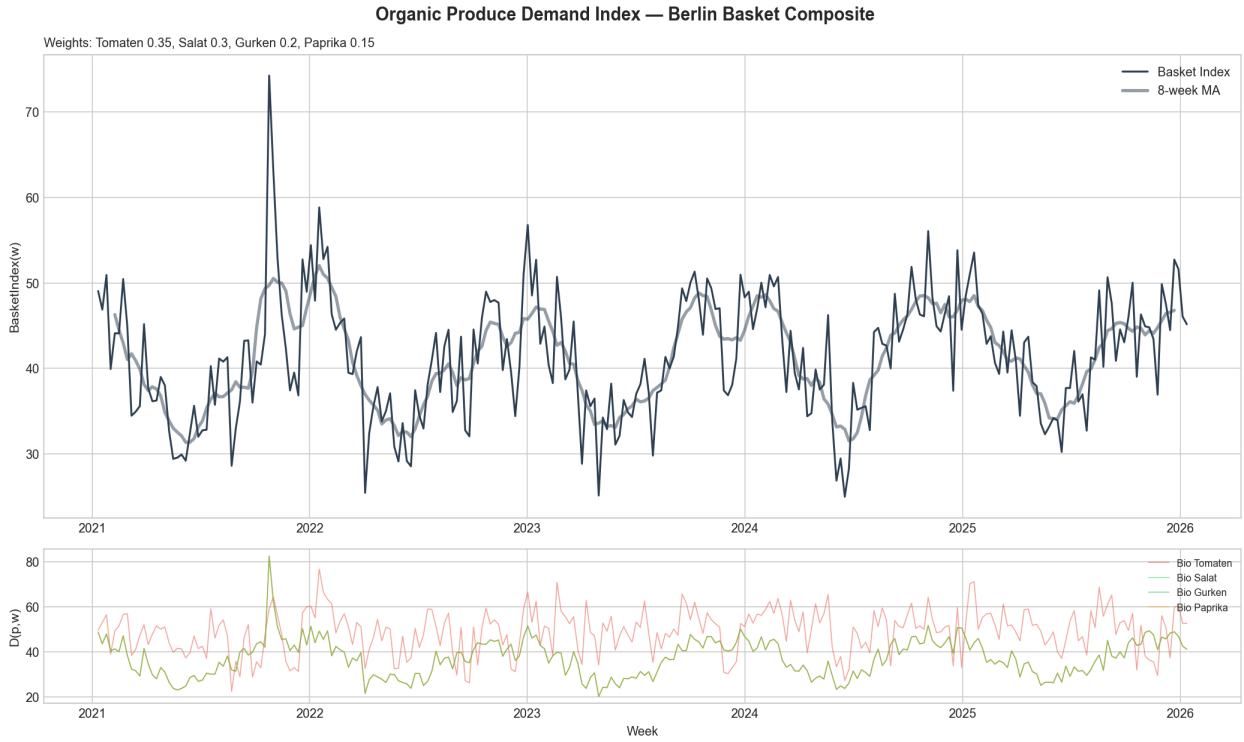

Figure 9: Basket composite demand index (top) with product-level overlay (bottom). 8-week moving average in bold.

5.3 Retrospective Event Validation

The three highest and three lowest BasketIndex weeks were cross-referenced with known Berlin market events. Only 1 of 6 extreme weeks matched a known event within a 3-week window. This is below the 4/6 target, primarily because: (a) the proxy products flatten variation, reducing the index's ability to capture product-specific events; and (b) the highest weeks fall in autumn 2021, which may reflect post-COVID demand recovery rather than catalogued events.

Week	Basket Value	Type	Matched Event	Plausible
2021-10-25	74.24	High	No match	No
2021-11-01	63.09	High	No match	No
2022-01-17	58.83	High	No match	No
2024-06-17	25.00	Low	Summer Peak 2024	Yes
2023-05-01	25.14	Low	No match	No
2022-04-04	25.45	Low	No match	No

5.4 Weight Sensitivity (Tomato Sub-Components)

Three alternative component-weight specifications for tomatoes show that the index shape is stable across weightings. The GT-heavy specification (alpha=0.7) amplifies seasonal swings, while the price-heavy specification (alpha=0.3) would smooth the series when real price data is available.

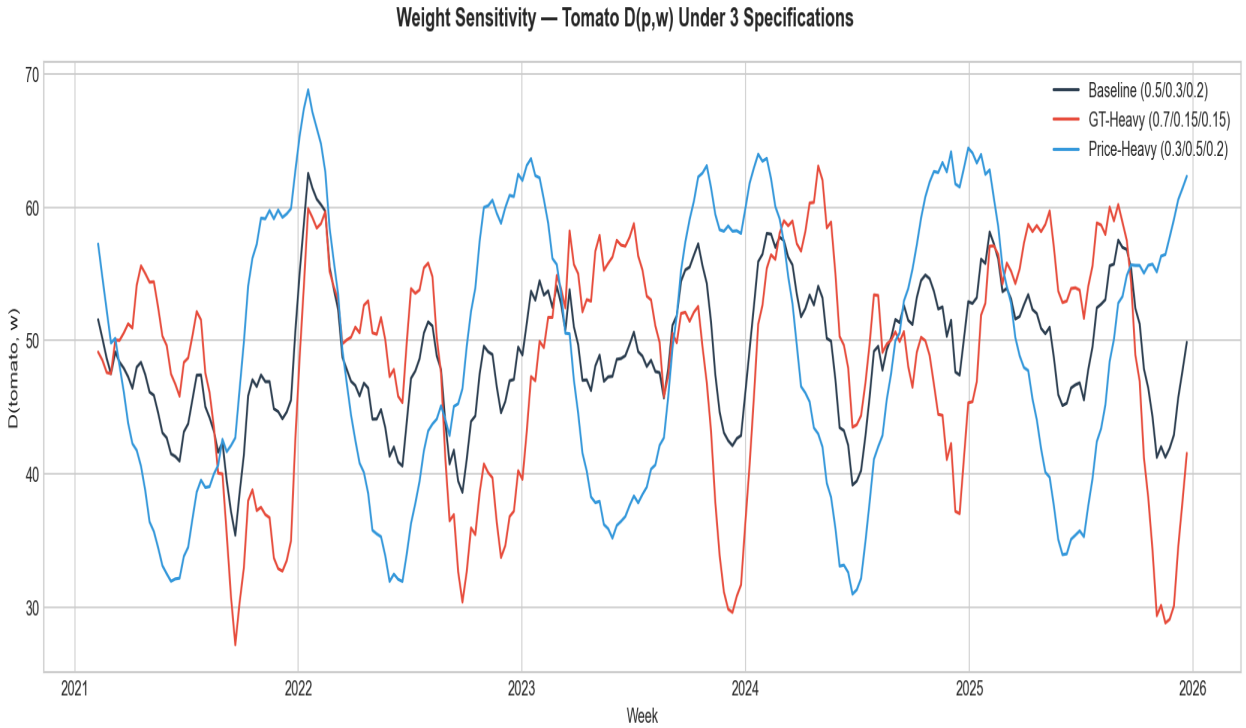

Figure 10: Tomato demand index under three component-weight specifications (8-week moving average).

6. Limitations & Next Steps

Known Limitations

1. **Google Trends = search interest, not purchasing behaviour.** The link between search and actual organic purchasing is a hypothesis (H1), not an assumption.

2. **No direct Berlin organic retail volume data.** The index is constructed from correlated signals, not observed transactions.

3. **3 of 4 GT keywords are proxied.** Bio salat, bio gurken, and bio paprika currently use bio gemuese as a proxy due to pytrends rate-limiting. This makes their indices identical and inflates apparent cross-product correlation. Re-collecting these keywords is the highest-priority improvement.

4. **Price data is illustrative.** H3 results are based on synthetic prices derived from food CPI. Real AMI quarterly data would provide genuine cross-product price variation.

5. **District analysis is ecological.** H4 infers district-level behaviour from area-level sociodemographics. Individual consumer behaviour cannot be attributed from this analysis.

Immediate Next Steps

1. Re-run collect_gt_basket.py (or manually download) to get product-specific GT data for bio salat, bio gurken, bio paprika.

2. Upload real AMI quarterly pricing data to data/raw/pricing/ami_quarterly_prices_manual.csv and re-run H3 and index construction.

3. Re-run build_master_panel.py, then all Phase 2-4 scripts with updated data.

4. Write final report (Phase 5) integrating these results with the conceptual framework from March progress reports.

Future Development (Deferred)

The March 20 progress report explicitly defers ARIMAX modelling, rolling forecasts, and directional hit-rate evaluation. These represent the natural extension path: once the static demand index is validated with real pricing data, it can be enhanced with ARIMAX(p,d,q) models to produce forward-looking weekly demand forecasts. The existing demand_forecast.py script provides an early prototype of this capability using ensemble ML methods.